

Bayesian Learning (Bayes Theorem)

→ Bayes rule / theorem helps to determine the probability of an event with random knowledge.

→ It is used to calculate the probability of occurring one event while other one already occurred.

→ It is a best method to relate the condition probability and marginal probability.

→ In simple words, we can say that Bayes theorem helps to contribute more accurate results.

~~→ It is hypo~~

→ Bayes theorem is one of the most popular machine learning concepts that helps to calculate the probability of occurring one event with uncertain knowledge while other one has already occurred.

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$P(A/B)$: conditional probability - Posterior Probability

$P(B/A)$: " " - likelihood

$P(A)$: Prior

$P(B)$: Marginal

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \quad (1), \quad P(B/A) = \frac{P(B \cap A)}{P(A)} \quad (2)$$

From equation (1)

$$P(A \cap B) = P(A/B) \cdot P(B) \quad (3)$$

From equation (2)

$$P(B \cap A) = P(B/A) \cdot P(A) \quad (4)$$

From equation (3) & (4), $P(A \cap B)$ and $P(B \cap A)$ are equal so,

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

why using Bayesian learning method in ML

- ↳ These methods are among most practical approaches to certain type learning.
- ↳ They provide useful perspective for understanding many learning algo. that do not explicitly manipulate probability.

Feature of Bayesian learning Methods:

- ↳ Instead of eliminating hypothesis, it will increase or decrease the estimated probability that hypothesis is correct.
- ↳ Prior knowledge can be combined with observed data to determine final probability of a hypothesis.
- ↳ This method provides hypothesis that makes probabilistic predictions.
- ↳ The new instance can be classified by combining the prediction of multiple hypotheses, weighted by their probabilities.
- ↳ This provides best stand decision making against which other practical methods can be measured, even Bayesian method passes computationally complex.

Issues

- 1) It requires initial knowledge of many probabilities. If it is not known in advance, they are estimated based on background knowledge, previous available data and assumption about the form of the underlying distribution.
2. Computational cost is high to determine Bayes optimal hypothesis in some special cases. It can be subsequently reduced.

Hypothesis

→ In machine learning we are often interested in determining the best hypothesis from some space H , given the observed training data D .

→ One way to specify what we mean by the best hypothesis is to say that we demand the most probable hypothesis, given the data D plus any initial knowledge about the prior probabilities of the various hypotheses in H .

→ Bayes theorem provides a direct method for calculate the probability of a hypothesis based on its prior probability, probabilities of observing various data given the hypothesis ~~based on its~~ ~~prior probability~~ and the observed data itself.

$$\rightarrow P(h/D) = \frac{P(D/h) \cdot P(h)}{P(D)}$$

$P(h)$ = probability that hypothesis h holds before we observed the training data. prior

$P(D)$ = prob. that training data D will be observed.

$P(D/h)$ = prob. of observing data D in which hypothesis h holds.

$P(h/D)$ = posterior probability. h holds after we have seen training data D .

~~$P(h/D) \uparrow$ unless $P(D/h), P(h)$ or~~

$P(h/D) \uparrow$ when $P(D/h) \uparrow$ or $P(h) \uparrow$ or $P(D) \downarrow$

$P(h/D) \downarrow$ as $P(D) \uparrow$

MAP (Maximum Posterior Hypothesis)

Given some set of candidate hypothesis H and interested in finding most probable hypothesis $h \in H$

Given the observed data D

Any such maximally probable hypothesis is called MAP

Calculating MAP using Bayes theorem

$$h_{MAP} = \operatorname{argmax}_{h \in H} P(h/D)$$

$$= \operatorname{argmax}_{h \in H} \frac{P(D/h) P(h)}{P(D)}$$

$$= \operatorname{argmax}_{h \in H} P(D/h) \cdot P(h)$$

We have dropped $P(D)$ as it is constant independent of h .

Maximum Likelihood (ML)

↳ Both ~~max~~ Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) are used to estimate parameters for a machine learning model.

↳ A special case of MAP where all the candidate hypothesis have same probability

$$\text{i.e. } P(h_i) = P(h_j)$$

In this case, we further simplify the equation

$$h_{ML} = \underset{h \in H}{\operatorname{argmax}} P(D/h)$$

where $P(D/h)$ = often called likelihood of data given h .

The hypothesis that maximizes $P(D/h)$ is called Maximum Likelihood hypothesis (h_{ML}).